

# BIG MART SALES PREDICTION SYSTEM

## Internship Project Report

### Submitted To

Naviotech Solution

### Submitted By

Abhishek Singh

### Project Title

Big Mart Sales Prediction Using Machine Learning

### Submission Date

May 2026

---

## CERTIFICATE

This is to certify that the project titled “**Big Mart Sales Prediction Using Machine Learning**” has been successfully completed by **Abhishek Singh** as part of the internship program requirement submitted to **Naviotech Solution**.

The work presented in this report is genuine and carried out under proper guidance and supervision.

### Project Guide

---

### Authorized Signature

---

---

## DECLARATION

I hereby declare that the project report entitled “**Big Mart Sales Prediction Using Machine Learning**” submitted in partial fulfillment of internship requirements is my original work.

The information and data included in this project are authentic to the best of my knowledge.

**Submitted By:**

Abhi Thakur

---

## ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Naviotech Solution for providing me with the opportunity to work on this project.

I would also like to thank my mentors, instructors, and colleagues for their continuous support and guidance throughout the completion of this project.

Finally, I thank my family and friends for their encouragement and motivation.

---

## ABSTRACT

Big Mart Sales Prediction is a machine learning-based project developed to predict the sales of products sold at various Big Mart stores. The main objective of this project is to analyze sales-related data and build a predictive model capable of estimating future sales.

The project uses data preprocessing, exploratory data analysis, feature engineering, and machine learning algorithms to achieve accurate predictions. Various factors such as item weight, item type, outlet size, outlet location, and item visibility are considered during model training.

Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn are used in the implementation. The Linear Regression algorithm is applied to train the prediction model.

This project demonstrates how machine learning can help businesses improve inventory management, increase sales efficiency, and make data-driven decisions.

---

## TABLE OF CONTENTS

1. Introduction
2. Problem Statement
3. Objectives
4. Scope of the Project
5. Technologies Used
6. Software & Hardware Requirements
7. Dataset Description
8. System Architecture
9. Flowchart

|                               |
|-------------------------------|
| 10. Methodology               |
| 11. Data Preprocessing        |
| 12. Exploratory Data Analysis |
| 13. Model Building            |
| 14. Source Code               |
| 15. Output Screenshots        |
| 16. Results & Discussion      |
| 17. Advantages                |
| 18. Limitations               |
| 19. Future Scope              |
| 20. Conclusion                |
| 21. References                |

---

## 1. INTRODUCTION

Sales prediction is one of the most important applications of machine learning in the retail industry. Businesses use sales forecasting to understand customer demand, optimize inventory, reduce losses, and improve profitability.

The Big Mart Sales Prediction System is designed to estimate product sales based on historical data. By analyzing multiple features related to products and store outlets, the system predicts the expected sales value.

Machine learning techniques are increasingly used in retail analytics because they can process large datasets efficiently and provide accurate predictions. This project demonstrates the implementation of a regression-based machine learning model for sales forecasting.

The project workflow includes:

- Data collection
- Data preprocessing
- Exploratory data analysis
- Feature encoding
- Model training
- Prediction and evaluation

This project is developed using Python programming language and machine learning libraries.

---

## 2. PROBLEM STATEMENT

Retail businesses face challenges in predicting future product sales due to changing customer behavior, seasonal demand, and varying store conditions.

Traditional forecasting methods are often inaccurate and time-consuming. Therefore, there is a need for an automated intelligent system that can analyze historical sales data and accurately predict future sales.

The problem addressed in this project is:

**“To develop a machine learning model capable of predicting Big Mart product sales using historical retail data.”**

---

### 3. OBJECTIVES

The major objectives of this project are:

- To analyze retail sales data.
  - To preprocess and clean the dataset.
  - To visualize important patterns in sales data.
  - To build a machine learning prediction model.
  - To evaluate model performance.
  - To predict future sales using regression techniques.
  - To understand the impact of product and outlet features on sales.
- 

### 4. SCOPE OF THE PROJECT

The scope of this project includes:

- Predicting sales performance.
- Helping retail businesses improve inventory management.
- Supporting data-driven decision-making.
- Reducing stock shortages and overstock situations.
- Enhancing customer satisfaction through better planning.

The project can be expanded in future by integrating advanced machine learning algorithms and real-time analytics.

---

### 5. TECHNOLOGIES USED

| Technology | Purpose               |
|------------|-----------------------|
| Python     | Programming Language  |
| NumPy      | Numerical Computation |

| Technology       | Purpose                   |
|------------------|---------------------------|
| Pandas           | Data Analysis             |
| Matplotlib       | Data Visualization        |
| Seaborn          | Statistical Visualization |
| Scikit-learn     | Machine Learning          |
| Jupyter Notebook | Development Environment   |

---

## 6. SOFTWARE & HARDWARE REQUIREMENTS

### Software Requirements

- Python 3.x
- Jupyter Notebook / Google Colab
- VS Code or PyCharm
- Windows/Linux Operating System

### Hardware Requirements

- Minimum 4GB RAM
- Intel i3 Processor or Higher
- 10GB Free Storage

---

## 7. DATASET DESCRIPTION

The dataset used in this project contains historical sales information of products sold across different Big Mart outlets.

### Important Features in Dataset

| Feature Name     | Description           |
|------------------|-----------------------|
| Item_Identifier  | Unique product ID     |
| Item_Weight      | Weight of product     |
| Item_Fat_Content | Fat content of item   |
| Item_Visibility  | Visibility percentage |
| Item_Type        | Product category      |

| Feature Name         | Description                   |
|----------------------|-------------------------------|
| Item_MRP             | Maximum Retail Price          |
| Outlet_Identifier    | Outlet ID                     |
| Outlet_Size          | Size of outlet                |
| Outlet_Location_Type | Location category             |
| Outlet_Type          | Type of store                 |
| Item_Outlet_Sales    | Sales value (Target Variable) |

## Dataset Characteristics

- Structured tabular dataset
- Contains numerical and categorical data
- Used for regression analysis

## 8. SYSTEM ARCHITECTURE

The architecture of the system consists of the following modules:

1. Data Collection Module
2. Data Preprocessing Module
3. Data Visualization Module
4. Feature Engineering Module
5. Model Training Module
6. Prediction Module
7. Result Evaluation Module

### Architecture Explanation

- Raw data is collected from the dataset.
- Data preprocessing removes missing values and converts categorical variables.
- Visualization helps identify trends and relationships.
- Machine learning model is trained using processed data.
- Predictions are generated and evaluated.

## 9. FLOWCHART

START

|

```
Load Dataset
|
Data Preprocessing
|
Exploratory Data Analysis
|
Feature Encoding
|
Train-Test Split
|
Model Training
|
Prediction
|
Performance Evaluation
|
END
```

---

## 10. METHODOLOGY

The methodology followed in this project is divided into multiple phases.

### Step 1: Data Collection

The dataset is loaded using Pandas.

### Step 2: Data Cleaning

Missing values and inconsistent data are handled.

### Step 3: Data Visualization

Graphs and charts are used to analyze patterns.

### Step 4: Feature Engineering

Categorical data is converted into numerical format using Label Encoding.

### Step 5: Model Training

The dataset is divided into training and testing sets.

## Step 6: Prediction

The Linear Regression algorithm predicts sales values.

## Step 7: Evaluation

Model performance is evaluated using regression metrics.

---

# 11. DATA PREPROCESSING

Data preprocessing is a very important step in machine learning.

The following preprocessing tasks were performed:

## Handling Missing Values

Missing values were checked and filled appropriately.

## Label Encoding

Categorical variables were transformed into numeric values.

Example:

```
from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()
train['Item_Fat_Content'] = encoder.fit_transform(train['Item_Fat_Content'])
```

## Feature Selection

Important features affecting sales prediction were selected.

## Train-Test Split

The dataset was split into:

- 80% Training Data
- 20% Testing Data

Example:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
```

---

## 12. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) helps understand patterns in data.

### Visualizations Used

#### 1. Distribution Plot

Used to understand sales distribution.

#### 2. Count Plot

Used to analyze categorical features.

#### 3. Heatmap

Used to understand feature correlation.

#### 4. Histogram

Used to analyze numerical features.

### Sample Visualization Code

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='Outlet_Size', data=train)
plt.show()
```

---

## 13. MODEL BUILDING

Machine learning model development is the core part of this project.

## Algorithm Used

### Linear Regression

Linear Regression is a supervised learning algorithm used for predicting continuous numerical values.

### Why Linear Regression?

- Simple and efficient
- Easy to implement
- Suitable for regression problems
- Fast training speed

### Model Training Code

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
```

### Prediction Code

```
prediction = model.predict(X_test)
```

---

## 14. SOURCE CODE

### Complete Python Code

```
# =====
# Big Mart Sales Prediction Project
# =====

# Import Required Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
```

```

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

import warnings
warnings.filterwarnings('ignore')

# =====
# Load Dataset
# =====

train = pd.read_csv('Train.csv')

# =====
# Display Dataset Information
# =====

print(train.head())
print(train.info())
print(train.describe())

# =====
# Check Missing Values
# =====

print(train.isnull().sum())

# Fill Missing Values
train['Item_Weight'].fillna(train['Item_Weight'].mean(), inplace=True)
train['Outlet_Size'].fillna(train['Outlet_Size'].mode()[0], inplace=True)

# =====
# Label Encoding
# =====

encoder = LabelEncoder()

categorical_columns = ['Item_Identifier', 'Item_Fat_Content', 'Item_Type',
                       'Outlet_Identifier', 'Outlet_Size',
                       'Outlet_Location_Type', 'Outlet_Type']

for col in categorical_columns:
    train[col] = encoder.fit_transform(train[col])

# =====
# Feature Selection
# =====

X = train.drop(columns='Item_Outlet_Sales', axis=1)

```

```

y = train['Item_Outlet_Sales']

# =====
# Train Test Split
# =====

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# =====
# Model Training
# =====

model = LinearRegression()
model.fit(X_train, y_train)

# =====
# Prediction
# =====

prediction = model.predict(X_test)

# =====
# Model Evaluation
# =====

mae = mean_absolute_error(y_test, prediction)
mse = mean_squared_error(y_test, prediction)
r2 = r2_score(y_test, prediction)

print("Mean Absolute Error:", mae)
print("Mean Squared Error:", mse)
print("R2 Score:", r2)

# =====
# Visualization
# =====

plt.figure(figsize=(8,6))
plt.scatter(y_test, prediction)
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title("Actual vs Predicted Sales")
plt.show()

```

## 15. OUTPUT SCREENSHOTS

### Output 1: Dataset Preview

- Displayed first few rows of dataset.
- Verified successful data loading.

### Output 2: Missing Values Check

- Identified missing values.
- Handled null data successfully.

### Output 3: Data Visualization Graphs

- Histograms
- Count plots
- Heatmaps

### Output 4: Model Training

- Linear Regression model trained successfully.

### Output 5: Prediction Results

- Predicted sales values generated.

### Output 6: Accuracy Metrics

- MAE
  - MSE
  - R2 Score
- 

## 16. RESULTS & DISCUSSION

The machine learning model successfully predicted sales values using historical retail data.

The Linear Regression model showed satisfactory performance for sales forecasting. Feature preprocessing and encoding improved the quality of predictions.

The following evaluation metrics were used:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

- R2 Score

The project demonstrates that machine learning can significantly help retail businesses in demand forecasting.

---

## 17. ADVANTAGES

- Improves sales forecasting accuracy.
  - Helps inventory management.
  - Reduces manual analysis.
  - Supports business decision-making.
  - Easy to implement and maintain.
  - Saves time and operational cost.
- 

## 18. LIMITATIONS

- Prediction accuracy depends on dataset quality.
  - Linear Regression may not capture complex patterns.
  - Real-time data integration is not included.
  - External market factors are not considered.
- 

## 19. FUTURE SCOPE

Future improvements can include:

- Using advanced algorithms like Random Forest and XGBoost.
  - Real-time sales prediction dashboards.
  - Web application deployment.
  - Integration with cloud platforms.
  - Deep learning-based forecasting.
  - Business intelligence reporting.
- 

## 20. CONCLUSION

The Big Mart Sales Prediction project successfully demonstrates the use of machine learning techniques for retail sales forecasting.

The project involved data preprocessing, visualization, feature engineering, model training, and prediction evaluation.

Linear Regression was used to build the prediction model, and the results showed that machine learning can effectively estimate future sales.

This project enhances practical understanding of machine learning concepts and their real-world business applications.

---

## 21. REFERENCES

1. Python Official Documentation
  2. Scikit-learn Documentation
  3. Pandas Documentation
  4. NumPy Documentation
  5. Matplotlib Documentation
  6. Seaborn Documentation
  7. Kaggle Big Mart Sales Dataset
- 

## APPENDIX

### Required Python Libraries Installation

```
pip install numpy
pip install pandas
pip install matplotlib
pip install seaborn
pip install scikit-learn
```

### Project Folder Structure

```
BigMartSalesPrediction/
|
|— dataset/
|   └─ Train.csv
|
|— notebooks/
|   └─ BigMartSalesPrediction.ipynb
|
|— screenshots/
|   └─ output1.png
|   └─ output2.png
|   └─ output3.png
```

```
|
|— report/
|   └─ Project_Report.pdf
|
|— presentation/
|   └─ Project_PPT.pptx
|
|— README.md
|
└─ requirements.txt
```

---

## README CONTENT FOR GITHUB

### Project Title

Big Mart Sales Prediction Using Machine Learning

### Description

This project predicts Big Mart product sales using machine learning algorithms.

### Features

- Data preprocessing
- Data visualization
- Machine learning model training
- Sales prediction
- Performance evaluation

### Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

### How to Run

```
python main.py
```

## **Author**

Abhi Thakur

---

**END OF REPORT**